

PRASHANT SHAH

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EDUCATION

Ramapo College of New Jersey

B.S. Data Science & Computer Science, Minor in Mathematics | GPA: 3.8/4.0

Recognitions: Best Research Award, ICPC NY Honorable Mention, Presidential Scholarship

Mahwah, NJ

May 2026

EXPERIENCE

Software Engineering Intern

May 2025 - Aug 2025

Trimble Inc.

Princeton, NJ

- Worked on AI-assistant workflow configuration in a C#/.NET enterprise platform, mapping which service data and payload fields moved in or out when triggered workflows fired across TMW Suite environments.
- Built REST APIs, SQL validation queries, and Azure/auth wiring so configuration changes became visible and editable before downstream assistant actions received stale or conflicting inputs.
- Used AWS S3/RDS, CloudWatch, Datadog traces, and SQL-backed checks to diagnose multi-environment inconsistencies and turn repeated support questions into reproducible validation paths.

Data Engineer (Contract)

Sep 2025 - Present

Center for Food Action

Mahwah, NJ

- Built an AI-assisted donation-intake workflow that classifies messy item names before save, routes uncertain records into review states, and blocks unverified PantrySoft handoff data before it leaves the system. Featured by Ramapo News and News12 NJ.
- Implemented Flask/PostgreSQL services, SQLAlchemy models, hooks, cron jobs, reconciliation queries, and reporting tables so staff could fix the intake process before automating exports into a third-party system.
- Used AI-assisted development loops to create documentation, test plans, and Playwright/API coverage around barcode scans, UPC lookup, donor sync, price fields, and review-before-save flows.

Research Assistant - Krag & Retrieval Systems (Link)

May 2025 - May 2026

Ramapo College of New Jersey

Mahwah, NJ

- Built and debugged an end-to-end RAG/Krag research pipeline on Vertex AI, constructing graph data from MovieLens 25M and orchestrating embedding, retrieval, and graph-model experiments across cloud runs.
- Researched content-intent retrieval where topical semantics and affective intent conflict; modeled User, Movie, and Emotion nodes with weighted EVOKES edges and Graph Transformer ranking aligned to Sentence-BERT.
- Used ablations and edge-removal tests to diagnose system behavior under dissonance, showing emotion edges were about 2.3x more necessary when signals conflicted.

Software Engineering Intern, Business Analytics

Jun 2024 - Aug 2024

Novartis Pharmaceuticals Corporation

East Hanover, NJ

- Built PostgreSQL data models and Azure Functions workflows to normalize project, budget, approval-status, and staffing data across 100+ internal initiatives in a regulated pharmaceutical environment.
- Delivered refreshable data-quality checks and Power BI/Tableau views that reduced a recurring 2-day clinical operations reporting workload while improving project and staffing visibility.

PROJECTS

Local Agent Workflow Architecture (Link)

- Built a Python/SQLite agent workflow cockpit where jobs, evidence, contacts, materials, statuses, and follow-ups are durable product state while Hermes runs as the API/tool engine behind streaming sessions.
- Integrated Hermes plugin tools, SSE runs, native tool dispatch, ATS hydration, contact caching, Gmail drafts, Typst/PDF compilation, provenance records, and external-action boundaries so agent work stays inspectable before upload or send.

Atlas LMS - course-record agent workspace (Link)

- Building a Next.js 16 / React 19 LMS where objectives, modules, assignments, submissions, proposals, and evidence share one org-scoped PostgreSQL/Drizzle course record across professor and student workflows.
- Implemented deterministic context packs, Vercel AI SDK typed block tools, Auth.js tenancy, BYOK provider keys, and proposal/apply transactions so agents can draft course changes without mutating live state.

SKILLS

Languages: Python, TypeScript, JavaScript, C#, Java, C/C++, SQL (PostgreSQL, MySQL, SQLite), Bash

AI & Agents: Claude Code, Codex, Cursor, AI SDKs, LLM APIs, tool calling, prompt engineering, stateful workflows

Backend/Cloud: .NET, Flask, FastAPI, Next.js, React, REST APIs, Azure Functions, AWS S3/RDS/CloudWatch, Docker

Quality/Data: Data structures, algorithms, system design basics, Playwright, pytest, Datadog traces, Vertex AI